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H2025-INF1321-01 - RESEAUX D'ENTREPRISE, SECURITE ET AUTOMATISATIONS

CHALLENGE LAB – ÉQUIPE

Timothe ALOGNON

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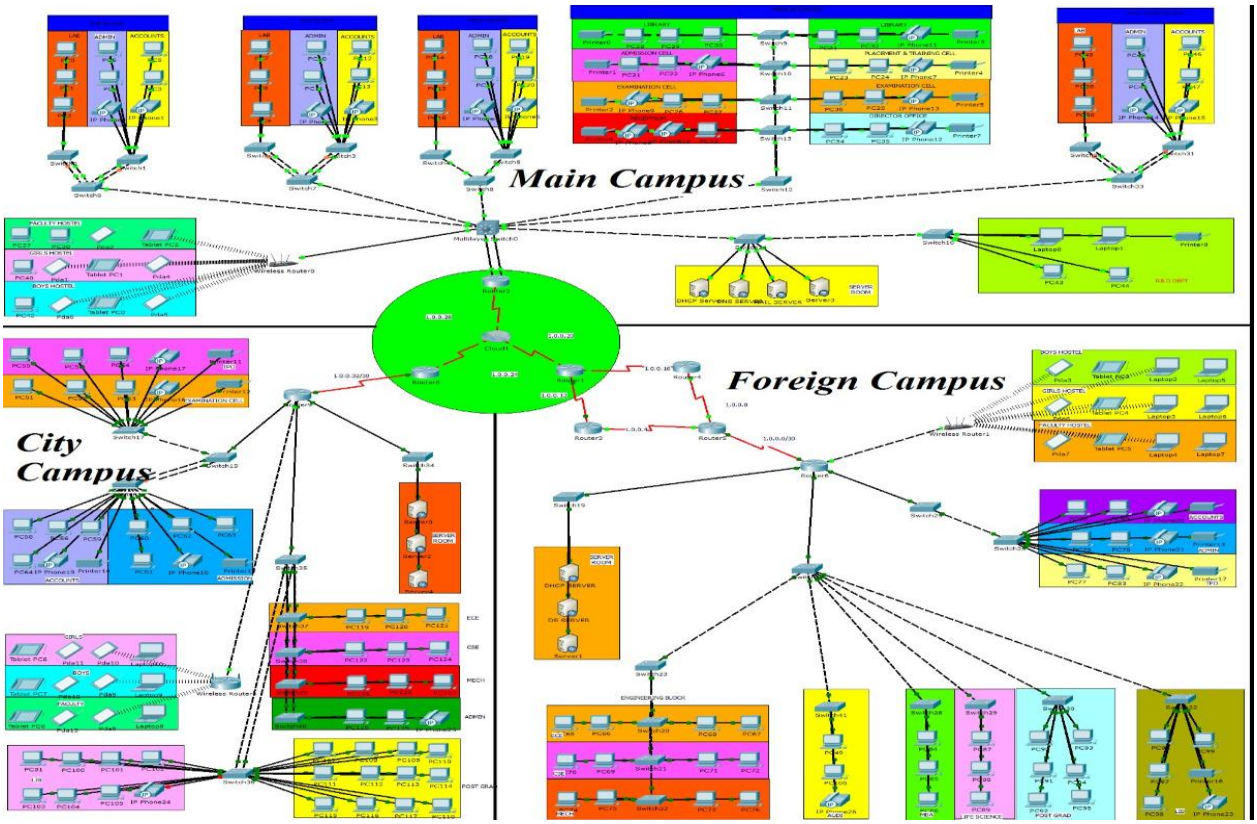
Partie 1 : Sous réseautage



Excellente illustration virtuelle d'un réseau **Multi-Campus,** cette topologie nous amène à construire et configurer un réseau complexe contenant un ensemble de protocole (OSPF**, CDP, LLDP)** assurant une connectivité fluide et sécurisée entre trois principaux sites dont Le **Foreign Campus** qui m'est attribué.

Ce site présente en son sein différents sous-réseaux (VLAN**)** et des équipements comme les **Switches** pour la gestion du routage inter-vlan, des **Routeurs** assurant la communication avec les autres campus, des **Points d'Accès Wifi** pour fournir une connectivité fluide et des serveurs et des téléphones IP.

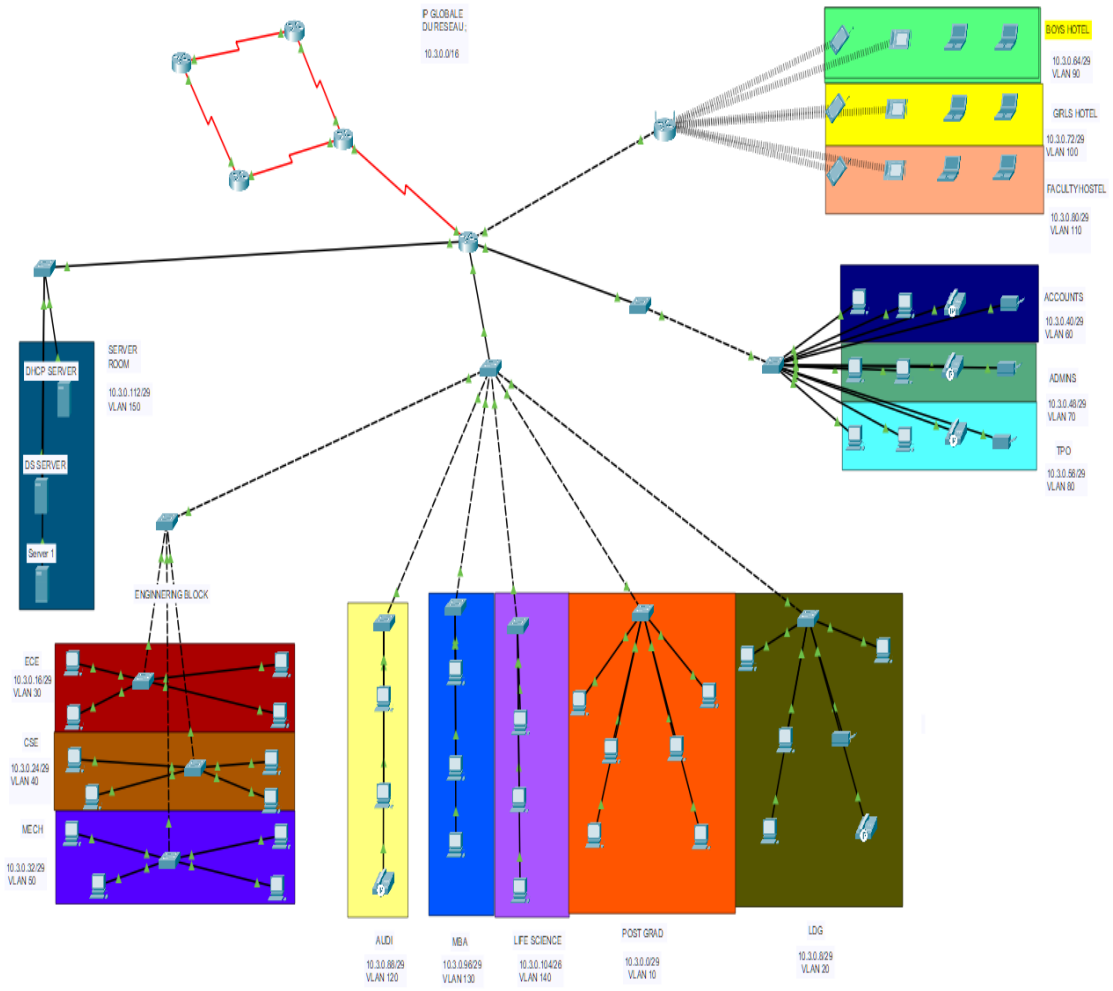
Le tableau suivant la structure **IP des sous-réseaux** de ma topologie :

Nom du sous-réseau	Adresse du sous-réseau	Masque du sous-réseau	Adresse du routeur	Dernière adresse utilisable	Adresse de diffusion	Nombre total de bits d'hôte	Nombre total d'hôtes
POST GRAD VLAN 10	10.3.0.0	255.255.255.248 (/29)	10.3.0.1	10.3.0.6	10.3.0.7	3	6
LDG VLAN 20	10.3.0.8	255.255.255.248 (/29)	10.3.0.9	10.3.0.14	10.3.0.15	3	6
ECE VLAN 30	10.3.0.16	255.255.255.248 (/29)	10.3.0.17	10.3.0.22	10.3.0.23	3	6
CSE VLAN 40	10.3.0.24	255.255.255.248 (/29)	10.3.0.25	10.3.0.30	10.3.0.31	3	6
MECH VLAN 50	10.3.0.32	255.255.255.248 (/29)	10.3.0.33	10.3.0.38	10.3.0.39	3	6
ACCOUNTS VLAN 60	10.3.0.40	255.255.255.248 (/29)	10.3.0.41	10.3.0.46	10.3.0.47	3	6
ADMINS VLAN 70	10.3.0.48	255.255.255.248 (/29)	10.3.0.49	10.3.0.54	10.3.0.55	3	6
TPO VLAN 80	10.3.0.56	255.255.255.248 (/29)	10.3.0.57	10.3.0.62	10.3.0.63	3	6
BOYS HOSTEL VLAN 90	10.3.0.64	255.255.255.248 (/29)	10.3.0.65	10.3.0.70	10.3.0.71	3	6
GIRLS HOSTEL VLAN 100	10.3.0.72	255.255.255.248 (/29)	10.3.0.73	10.3.0.78	10.3.0.79	3	6
FACULTY HOSTEL VLAN 110	10.3.0.80	255.255.255.248 (/29)	10.3.0.81	10.3.0.86	10.3.0.87	3	6

AUDI VLAN 120	**10.3.0.88**	**255.255.255.248 (/29)**	**10.3.0.89**	**10.3.0.94**	**10.3.0.95**	**3**	**6**
MBA VLAN 130	**10.3.0.96**	**255.255.255.248 (/29)**	**10.3.0.97**	**10.3.0.102**	**10.3.0.103**	**3**	**6**
LIFE SCIENCE VLAN 140	**10.3.0.104**	**255.255.255.248 (/29)**	**10.3.0.105**	**10.3.0.110**	**10.3.0.111**	**3**	**6**
SERVER ROOM VLAN 150	**10.3.0.112**	**255.255.255.248 (/29)**	**10.3.0.113**	**10.3.0.118**	**10.3.0.119**	**3**	**6**

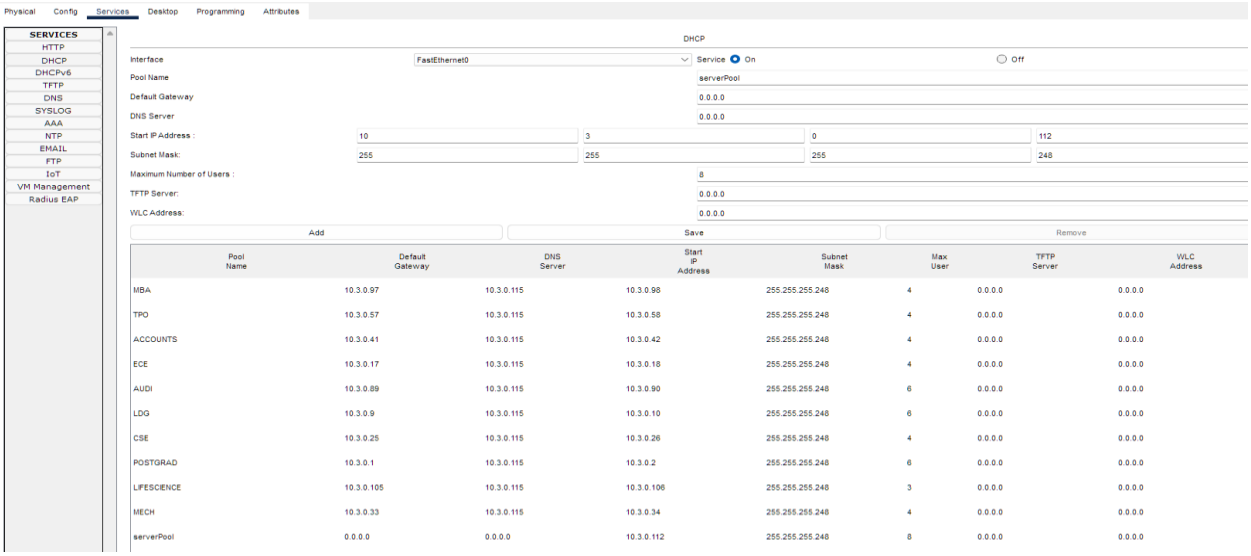
Partie 2 – Branchement physique et le choix des équipements

Dans une conception réseau le choix de bons équipements est un facteur très important. Dans ma topologie j'ai eu à utiliser des **Switches Virtuels C2960-24TT pour la gestion des trunks,** des **Routeurs Virtuel 2811 pour la configuration du routage (Router on Stick), un Access Point WRT300N, des téléphones IP 7960, des Laptop, des PC et des Printer-PT comme le présente la suivante topologie :**

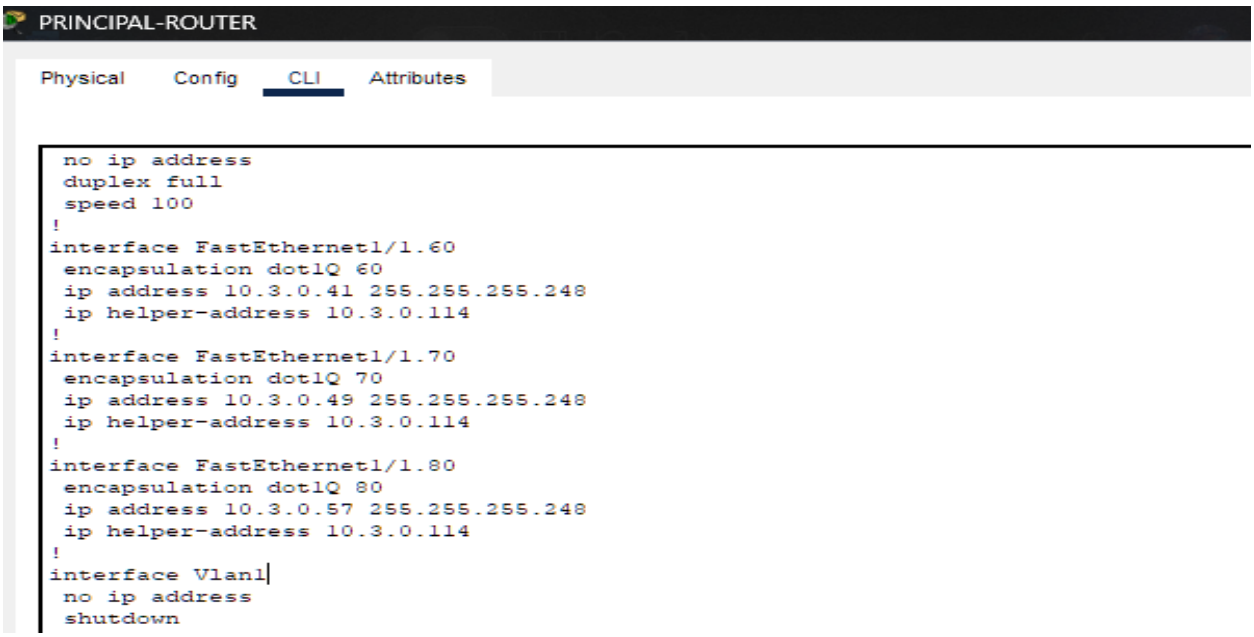


Partie 3 – Configuration des équipements finaux (ordi, Imp, tel, etc.)

Nous allons procéder dans cette partie à la **Configuration IP des équipements.** Pour cela nous allons mettre en place un server **DHCP pour les équipements comme les PC, Printers etc…. mais une configuration statique pour les routeurs et Accès Point.**



Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
MBA	10.3.0.97	10.3.0.115	10.3.0.98	255.255.255.248	4	0.0.0.0	0.0.0.0
TPO	10.3.0.57	10.3.0.115	10.3.0.58	255.255.255.248	4	0.0.0.0	0.0.0.0
ACCOUNTS	10.3.0.41	10.3.0.115	10.3.0.42	255.255.255.248	4	0.0.0.0	0.0.0.0
ECE	10.3.0.17	10.3.0.115	10.3.0.18	255.255.255.248	4	0.0.0.0	0.0.0.0
AUDI	10.3.0.89	10.3.0.115	10.3.0.90	255.255.255.248	6	0.0.0.0	0.0.0.0
LDG	10.3.0.9	10.3.0.115	10.3.0.10	255.255.255.248	6	0.0.0.0	0.0.0.0
CSE	10.3.0.25	10.3.0.115	10.3.0.26	255.255.255.248	4	0.0.0.0	0.0.0.0
POSTGRAD	10.3.0.1	10.3.0.115	10.3.0.2	255.255.255.248	6	0.0.0.0	0.0.0.0
LIFESCIENCE	10.3.0.105	10.3.0.115	10.3.0.106	255.255.255.248	3	0.0.0.0	0.0.0.0
MECH	10.3.0.33	10.3.0.115	10.3.0.34	255.255.255.248	4	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	10.3.0.112	255.255.255.248	8	0.0.0.0	0.0.0.0




```
 no ip address
 duplex full
 speed 100
!
interface FastEthernet1/1.60
 encapsulation dot1Q 60
 ip address 10.3.0.41 255.255.255.248
 ip helper-address 10.3.0.114
!
interface FastEthernet1/1.70
 encapsulation dot1Q 70
 ip address 10.3.0.49 255.255.255.248
 ip helper-address 10.3.0.114
!
interface FastEthernet1/1.80
 encapsulation dot1Q 80
 ip address 10.3.0.57 255.255.255.248
 ip helper-address 10.3.0.114
!
interface Vlan1
 no ip address
 shutdown
```

PRINCIPAL-ROUTER

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
!
!
!
!
interface FastEthernet0/0
 ip address 10.3.0.113 255.255.255.240
 duplex full
 speed 100
!
interface FastEthernet0/1
 no ip address
 duplex full
 speed 100
!
interface FastEthernet0/1.10
 encapsulation dot1Q 10
 ip address 10.3.0.1 255.255.255.240
 ip helper-address 10.3.0.114
!
interface FastEthernet0/1.20
 encapsulation dot1Q 20
 ip address 10.3.0.9 255.255.255.240
 ip helper-address 10.3.0.114
!
interface FastEthernet0/1.30
 encapsulation dot1Q 30
 ip address 10.3.0.17 255.255.255.240
 ip helper-address 10.3.0.114
!
interface FastEthernet0/1.40
 encapsulation dot1Q 40
 ip address 10.3.0.25 255.255.255.240
 ip helper-address 10.3.0.114
!
interface FastEthernet0/1.50
 encapsulation dot1Q 50
 ip address 10.3.0.33 255.255.255.240
 ip helper-address 10.3.0.114
!
interface FastEthernet0/1.120
 encapsulation dot1Q 120
 ip address 10.3.0.89 255.255.255.240
 ip helper-address 10.3.0.114
!
interface FastEthernet0/1.130
 encapsulation dot1Q 130
 ip address 10.30.0.97 255.255.255.240
 ip helper-address 10.3.0.114
!
interface FastEthernet0/1.140
 encapsulation dot1Q 140
 ip address 10.3.0.105 255.255.255.240
 ip helper-address 10.3.0.114
```

PC19

Physical | Config | Desktop | Programming | Attributes | pt-sm://net.netacad.cisco.activitysequencer/index.html

IP Configuration

Interface: FastEthernet0

IP Configuration

DHCP | Static

IPv4 Address	10.3.0.20
Subnet Mask	255.255.255.248
Default Gateway	10.3.0.17
DNS Server	10.3.0.115

PC21

Physical | Config | Desktop | Programming | Attributes | pt-sm://net.netacad.cisco.activitysequencer/index.html

IP Configuration

Interface: FastEthernet0

IP Configuration

DHCP | Static

IPv4 Address	10.3.0.10
Subnet Mask	255.255.255.248
Default Gateway	10.3.0.9
DNS Server	10.3.0.115

IPv6 Configuration

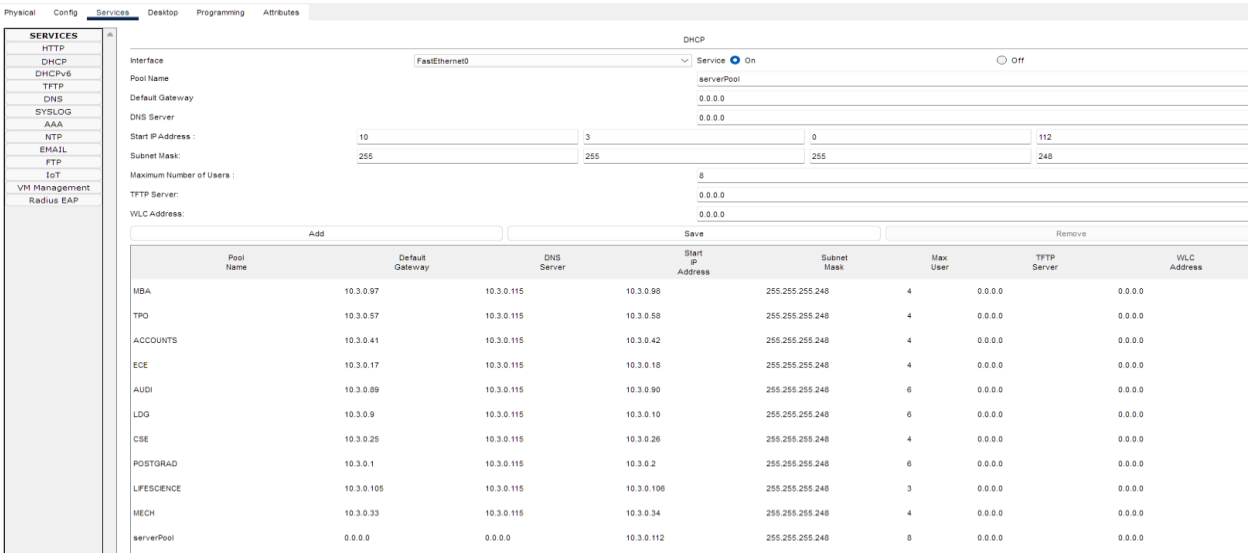
Partie 4 – Configuration des commutateurs et routeurs

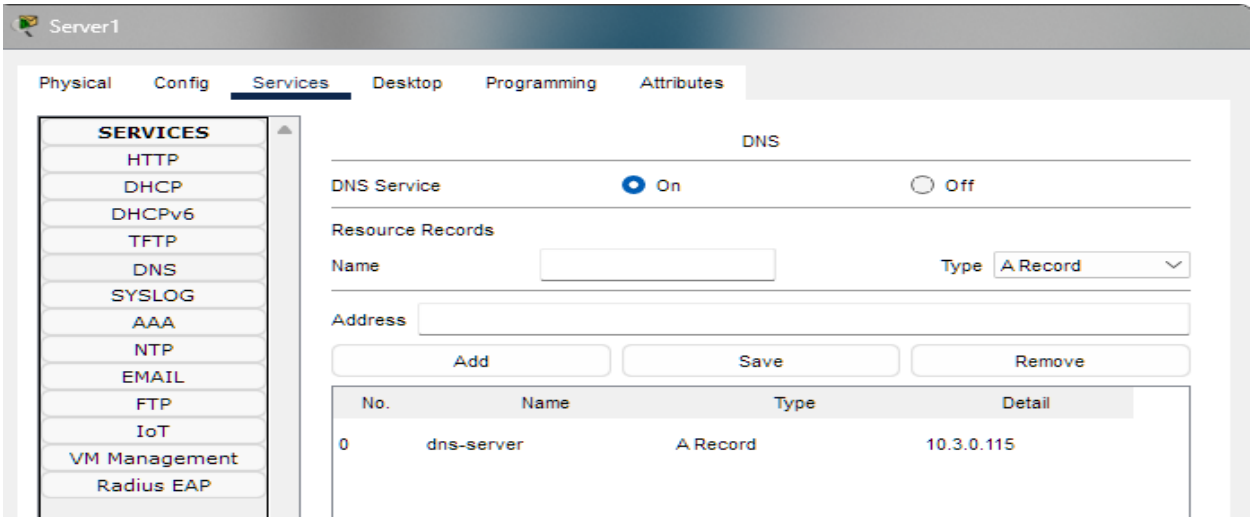
Dans cette partie je vais procéder à la sécurité de nos équipements par des actions et configurations comme **un accès exécutif et utilisateur sécurisé par des mots de passes ; la mise en place du protocole CDP et LLDP pour la reconnaissance des voisins connectés à un équipement ; un accès distant SSH sécurisé ; mise en shutdown des interfaces non utilisés, définition de la vitesse et du mode duplex sur les interfaces nécessaires. Ces suivantes captures d'écran montrent l'aperçu des configurations :**

```
PRINCIPAL-SWITCH#sh ru
PRINCIPAL-SWITCH#sh running-config
Building configuration...

Current configuration : 1941 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname PRINCIPAL-SWITCH
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
ip ssh version 2
ip domain-name kbu.local
!
!
!
lldp run
!
!
line con 0
 password 7 0822404F1A0A
 login
!
line vty 0 4
 login local
 transport input ssh
line vty 5 15
 login local
 transport input ssh
!
!
!
!
end


PRINCIPAL-SWITCH#
```

```
PRINCIPAL-ROUTER#sh cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID     Local Intrfce    Holdtme    Capability    Platform    Port ID
SERVER-SWITCH
              Fas 0/0          151            S         2960        Fas 0/4
Router        Ser 0/2/0        150            R         C1841       Ser 0/0/0
PRINCIPAL-ROUTER#
```

```
PRINCIPAL-ROUTER#sh lldp neighbors
Capability codes:
    (R) Router, (B) Bridge, (T) Telephone, (C) DOCSIS Cable Device
    (W) WLAN Access Point, (P) Repeater, (S) Station, (O) Other
Device ID           Local Intf     Hold-time  Capability      Port ID
PRINCIPAL-SWITCH    Fa             120        B               Fa0/2
ACCOUNTS-ADMINS-TPO-PRINCIPALFa              120        B               Fa0/18
SERVER-SWITCH       Fa0/0          120        B               Fa0/4

Total entries displayed: 3
PRINCIPAL-ROUTER#
```





Partie 5 – Rapport de L'examen

Je présente ici les résultats des commandes importantes comme **show run ; show ip int br ; show ip route ; show vlan sur les principaux équipements :**

ROUTEURS

```
PRINCIPAL-ROUTER#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.3.0.113      YES manual up                    up
FastEthernet0/1        unassigned      YES unset  up                    up
FastEthernet0/1.10     10.3.0.1        YES manual up                    up
FastEthernet0/1.20     10.3.0.9        YES manual up                    up
FastEthernet0/1.30     10.3.0.17       YES manual up                    up
FastEthernet0/1.40     10.3.0.25       YES manual up                    up
FastEthernet0/1.50     10.3.0.33       YES manual up                    up
FastEthernet0/1.120    10.3.0.89       YES manual up                    up
FastEthernet0/1.130    10.30.0.97      YES manual up                    up
FastEthernet0/1.140    10.3.0.105      YES manual up                    up
Serial0/2/0            unassigned      YES unset  up                    up
Serial0/2/1            unassigned      YES unset  administratively down down
FastEthernet1/0        unassigned      YES unset  up                    up
FastEthernet1/1        unassigned      YES unset  up                    up
FastEthernet1/1.60     10.3.0.41       YES manual up                    up
FastEthernet1/1.70     10.3.0.49       YES manual up                    up
FastEthernet1/1.80     10.3.0.57       YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

```
PRINCIPAL-ROUTER#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 24 subnets, 2 masks
C       10.3.0.0/29 is directly connected, FastEthernet0/1.10
L       10.3.0.1/32 is directly connected, FastEthernet0/1.10
C       10.3.0.8/29 is directly connected, FastEthernet0/1.20
L       10.3.0.9/32 is directly connected, FastEthernet0/1.20
C       10.3.0.16/29 is directly connected, FastEthernet0/1.30
L       10.3.0.17/32 is directly connected, FastEthernet0/1.30
C       10.3.0.24/29 is directly connected, FastEthernet0/1.40
L       10.3.0.25/32 is directly connected, FastEthernet0/1.40
C       10.3.0.32/29 is directly connected, FastEthernet0/1.50
L       10.3.0.33/32 is directly connected, FastEthernet0/1.50
C       10.3.0.40/29 is directly connected, FastEthernet1/1.60
L       10.3.0.41/32 is directly connected, FastEthernet1/1.60
C       10.3.0.48/29 is directly connected, FastEthernet1/1.70
L       10.3.0.49/32 is directly connected, FastEthernet1/1.70
C       10.3.0.56/29 is directly connected, FastEthernet1/1.80
L       10.3.0.57/32 is directly connected, FastEthernet1/1.80
C       10.3.0.88/29 is directly connected, FastEthernet0/1.120
L       10.3.0.89/32 is directly connected, FastEthernet0/1.120
C       10.3.0.104/29 is directly connected, FastEthernet0/1.140
L       10.3.0.105/32 is directly connected, FastEthernet0/1.140
C       10.3.0.112/29 is directly connected, FastEthernet0/0
L       10.3.0.113/32 is directly connected, FastEthernet0/0
C       10.30.0.96/29 is directly connected, FastEthernet0/1.130
L       10.30.0.97/32 is directly connected, FastEthernet0/1.130
```

```
PRINCIPAL-ROUTER#sh running-config
Building configuration...

Current configuration: 2439 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname PRINCIPAL-ROUTER
!
!
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2811/K9 sn FTX1017HNQ0-
!
!
!
lldp run
!
!
!
!
!
!
!
ip ssh version 2
ip domain-name kbu. local
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface FastEthernet0/0
ip address 10.3.0.113 255.255.255.248
duplex full
speed 100
!
interface FastEthernet0/1
no ip address
duplex full
speed 100
!
interface FastEthernet0/1.10
encapsulation dot1Q 10
ip address 10.3.0.1 255.255.255.248
ip helper-address 10.3.0.114
!
interface FastEthernet0/1.20
```

```
encapsulation dot1Q 20
ip address 10.3.0.9 255.255.255.248
ip helper-address 10.3.0.114
!
interface FastEthernet0/1.30
encapsulation dot1Q 30
ip address 10.3.0.17 255.255.255.248
ip helper-address 10.3.0.114
!
interface FastEthernet0/1.40
encapsulation dot1Q 40
ip address 10.3.0.25 255.255.255.248
ip helper-address 10.3.0.114
!
interface FastEthernet0/1.50
encapsulation dot1Q 50
ip address 10.3.0.33 255.255.255.248
ip helper-address 10.3.0.114
!
interface FastEthernet0/1.120
encapsulation dot1Q 120
ip address 10.3.0.89 255.255.255.248
ip helper-address 10.3.0.114
!
interface FastEthernet0/1.130
encapsulation dot1Q 130
ip address 10.30.0.97 255.255.255.248
ip helper-address 10.3.0.114
!
interface FastEthernet0/1.140
encapsulation dot1Q 140
ip address 10.3.0.105 255.255.255.248
ip helper-address 10.3.0.114
!
interface Serial0/2/0
no ip address
clock rate 2000000
!
interface Serial0/2/1
no ip address
clock rate 2000000
shutdown
!
interface FastEthernet1/0
no ip address
duplex auto
speed auto
!
interface FastEthernet1/1
no ip address
duplex full
speed 100
!
interface FastEthernet1/1.60
encapsulation dot1Q 60
ip address 10.3.0.41 255.255.255.248
ip helper-address 10.3.0.114
!
interface FastEthernet1/1.70
encapsulation dot1Q 70
ip address 10.3.0.49 255.255.255.248
ip helper-address 10.3.0.114
!
interface FastEthernet1/1.80
encapsulation dot1Q 80
ip address 10.3.0.57 255.255.255.248
```

```
ip helper-address 10.3.0.114
!
interface Vlan1
no ip address
shutdown
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
password class
login
!
line aux 0
!
line vty 0 4
login local
transport input ssh
line vty 5 15
login local
transport input ssh
!
!
!
end
PRINCIPAL-ROUTER#
```

SWITCHS

PRINCIPAL-SWITCH

```
PRINCIPAL-SWITCH#sh vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Gig0/1, Gig0/2
10   POSTGRAB                         active    Fa0/24
20   LDG                              active
30   ECE                              active
40   CSE                              active
50   MECH                             active
60   ACCOUNTS                         active
70   ADMINS                           active
80   TPO                              active
120  AUDI                             active
130  MBA                              active
140  LIFESCIENCE                      active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
10   enet  100010     1500  -      -      -        -    -        0      0
20   enet  100020     1500  -      -      -        -    -        0      0
30   enet  100030     1500  -      -      -        -    -        0      0
40   enet  100040     1500  -      -      -        -    -        0      0
50   enet  100050     1500  -      -      -        -    -        0      0
60   enet  100060     1500  -      -      -        -    -        0      0
70   enet  100070     1500  -      -      -        -    -        0      0
80   enet  100080     1500  -      -      -        -    -        0      0
120  enet  100120     1500  -      -      -        -    -        0      0
130  enet  100130     1500  -      -      -        -    -        0      0
140  enet  100140     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------

Remote SPAN VLANs
------------------------------------------------------------------------------

Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------
```

```
PRINCIPAL-SWITCH#show ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual down                  down
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual down                  down
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual up                    up
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual up                    up
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down
FastEthernet0/22       unassigned      YES manual down                  down
FastEthernet0/23       unassigned      YES manual down                  down
FastEthernet0/24       unassigned      YES manual down                  down
GigabitEthernet0/1     unassigned      YES manual down                  down
GigabitEthernet0/2     unassigned      YES manual down                  down
Vlan1                  unassigned      YES manual administratively down down
PRINCIPAL-SWITCH#
```

```
PRINCIPAL-SWITCH#sh running-config
Building configuration...

Current configuration: 1998 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname PRINCIPAL-SWITCH
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
ip ssh version 2
ip domain-name kbu. local
!
username admin secret 5 $1$mERr$vTbHul1N28cEp8lkLqr0f/
!
!
!
lldp run
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
switchport trunk allowed vlan 20
switchport mode trunk
!
interface FastEthernet0/2
switchport trunk allowed vlan 10,20,30,40,50,120,130,140
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet0/3
switchport trunk allowed vlan 10
switchport mode trunk
!
interface FastEthernet0/4
switchport trunk allowed vlan 130
switchport mode trunk
!
interface FastEthernet0/5
switchport trunk allowed vlan 120
switchport mode trunk
!
interface FastEthernet0/6
switchport trunk allowed vlan 30,40,50
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet0/7
!
```

```
interface FastEthernet0/8
!
interface FastEthernet0/9
switchport trunk allowed vlan 140
switchport mode trunk
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
switchport trunk allowed vlan 10,20,120,130,140
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
switchport access vlan 10
switchport trunk allowed vlan 60,70,80
switchport mode trunk
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
no ip address
shutdown
!
!
!
!
line con 0
password 7 0822404F1A0A
login
!
```

```
line vty 0 4
login local
transport input ssh
line vty 5 15
login local
transport input ssh
!
!
!
!
end
PRINCIPAL-SWITCH#
```

ENGINEERING BLOCK

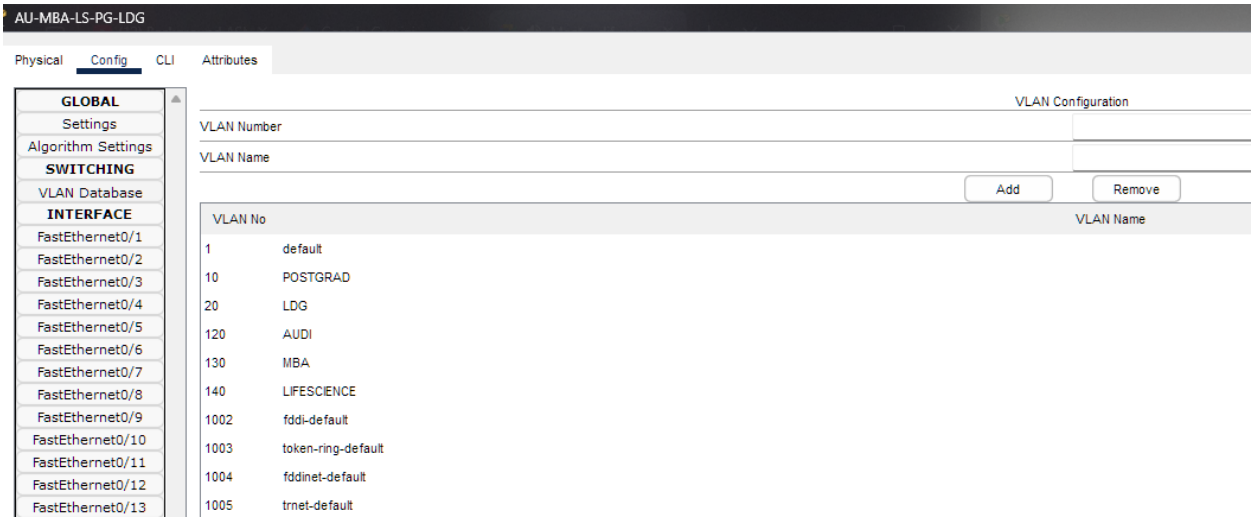
```
ENGINEERING-BLOCK#sh vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Gig0/1, Gig0/2
30   ECE                              active
40   CSE                              active
50   MECH                             active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
30   enet  100030     1500  -      -      -        -    -        0      0
40   enet  100040     1500  -      -      -        -    -        0      0
50   enet  100050     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------

Remote SPAN VLANs
------------------------------------------------------------------------------

Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------
ENGINEERING-BLOCK#
```

```
ENGINEERING-BLOCK#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual down                  down
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual down                  down
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual up                    up
FastEthernet0/7        unassigned      YES manual up                    up
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down
FastEthernet0/22       unassigned      YES manual down                  down
FastEthernet0/23       unassigned      YES manual down                  down
FastEthernet0/24       unassigned      YES manual up                    up
GigabitEthernet0/1     unassigned      YES manual down                  down
GigabitEthernet0/2     unassigned      YES manual down                  down
Vlan1                  unassigned      YES manual administratively down down
ENGINEERING-BLOCK#
```

```
ENGINEERING-BLOCK#sh running-config
Building configuration...

Current configuration: 1600 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname ENGINEERING-BLOCK
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
ip ssh version 2
ip domain-name kbu. local
!
!
!
lldp run
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2
switchport trunk allowed vlan 30
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
switchport trunk allowed vlan 40
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet0/7
switchport trunk allowed vlan 50
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
```

```
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
switchport trunk allowed vlan 30,40,50
switchport mode trunk
duplex full
speed 100
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
no ip address
shutdown
!
!
!
!
line con 0
password class
login
!
line vty 0 4
login local
transport input ssh
line vty 5 15
login local
transport input ssh
!
!
!
!
End
ENGINEERING-BLOCK#
```

AU-MBA-LS-PG-LDG



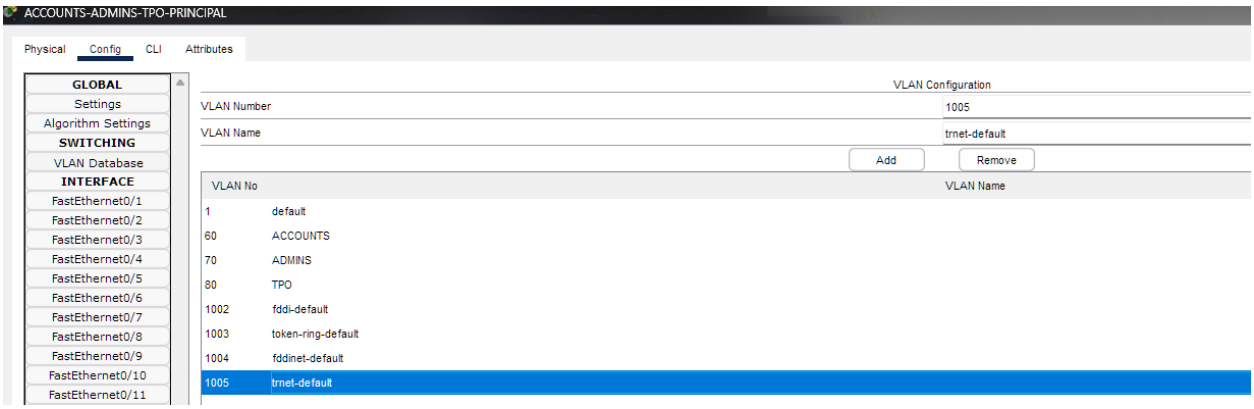

```
AU-MBA-LS-PG-LDG#sh ip int brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual down                  down
FastEthernet0/2        unassigned      YES manual down                  down
FastEthernet0/3        unassigned      YES manual down                  down
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual up                    up
FastEthernet0/20       unassigned      YES manual up                    up
FastEthernet0/21       unassigned      YES manual up                    up
FastEthernet0/22       unassigned      YES manual up                    up
FastEthernet0/23       unassigned      YES manual up                    up
FastEthernet0/24       unassigned      YES manual up                    up
GigabitEthernet0/1     unassigned      YES manual down                  down
GigabitEthernet0/2     unassigned      YES manual down                  down
Vlan1                  unassigned      YES manual administratively down down
AU-MBA-LS-PG-LDG#
```

```
 AU-MBA-LS-PG-LDG#sh running-config
Building configuration...

Current configuration: 1773 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
Hostname AU-MBA-LS-PG-LDG
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
ip ssh version 2
ip domain-name kbu. local
!
!
!
lldp run
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
```

```
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
switchport trunk allowed vlan 10,20,120,130,140
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet0/20
switchport trunk allowed vlan 20
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet0/21
switchport trunk allowed vlan 10
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet0/22
switchport trunk allowed vlan 140
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet0/23
switchport trunk allowed vlan 130
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet0/24
switchport trunk allowed vlan 120
switchport mode trunk
duplex full
speed 100
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
no ip address
shutdown
!
!
!
!
line con 0
password class
login
!
line vty 0 4
login local
transport input ssh
```

```
line vty 5 15
login local
transport input ssh
!
!
!
!
end
AU-MBA-LS-PG-LDG#
```

ACCOUNTS-ADMINS-TPO-PRINCIPAL




```
ACCOUNTS-ADMINS-TPO-PRINCIPAL#
%SYS-5-CONFIG_I: Configured from console by console

ACCOUNTS-ADMINS-TPO-PRINCIPAL#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual down                  down
FastEthernet0/2        unassigned      YES manual down                  down
FastEthernet0/3        unassigned      YES manual down                  down
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual up                    up
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down
FastEthernet0/22       unassigned      YES manual down                  down
FastEthernet0/23       unassigned      YES manual down                  down
FastEthernet0/24       unassigned      YES manual up                    up
GigabitEthernet0/1     unassigned      YES manual down                  down
GigabitEthernet0/2     unassigned      YES manual down                  down
Vlan1                  unassigned      YES manual administratively down down
ACCOUNTS-ADMINS-TPO-PRINCIPAL#
```

```
ACCOUNTS-ADMINS-TPO-PRINCIPAL#sh running-config
Building configuration...

Current configuration : 1432 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname ACCOUNTS-ADMINS-TPO-PRINCIPAL
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
ip ssh version 2
ip domain-name kbu. local
!
!
!
lldp run
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
```

```
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
switchport trunk allowed vlan 60,70,80
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
switchport trunk allowed vlan 60,70,80
switchport mode trunk
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
no ip address
shutdown
!
!
!
!
line con 0
password class
login
!
line vty 0 4
login local
transport input ssh
line vty 5 15
login local
transport input ssh
!
!
!
!
end
```

TEST DE PING

vLAN ECE vers VLAN CSE

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::201:64FF:FE66:E0A6
   IPv6 Address....................: ::
   IPv4 Address....................: 10.3.0.19
   Subnet Mask.....................: 255.255.255.248
   Default Gateway.................: ::
                                     10.3.0.17

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 10.3.0.29

Pinging 10.3.0.29 with 32 bytes of data:

Reply from 10.3.0.29: bytes=32 time<1ms TTL=127
Reply from 10.3.0.29: bytes=32 time<1ms TTL=127
Reply from 10.3.0.29: bytes=32 time<1ms TTL=127
Reply from 10.3.0.29: bytes=32 time<1ms TTL=127

Ping statistics for 10.3.0.29:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Partie 6 – Conclusion

Avec des difficultés au début ce projet m'a encore une fois amené à approfondir et bien tenir en main mes compétences dans la configuration et la sécurité réseau. Il a été un peu complexe pour moi surtout lors de la phase de mise en place des sous-réseaux et de la configuration du serveur DHCP mais il reste encore plein de chose que je pourrais améliorer pour la sécurité comme **la restriction du service DHCP au Mac-address**